

On the triviality of inhomogeneous deformations of $\mathfrak{osp}(1|2n)$

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Abstract

We specify a symmetrized mixed-oscillator deformation family of $B(0, n) = \mathfrak{osp}(1|2n)$, with even mixed coefficients and one odd square-zero parameter. For every $n \geq 1$, we derive its bracket from a faithful oscillator realization and exhibit an odd cochain whose coboundary is the recovered deformation coefficient. The resulting even change of generators is an exact isomorphism over the exterior parameter algebra. For $n = 1$, the cochain agrees with the normalization of Bakalov–Sullivan. We give the source relations and the even-central specialization explicitly, together with a Lean 4 formalization.

1 Introduction

Bakalov and Sullivan introduce, in [1], a theory of supersymmetric bilinear forms $(\cdot|\cdot)$ on a super-space $V = V_{\bar{0}} \oplus V_{\bar{1}}$ that they call *inhomogeneous* when $(V_{\bar{0}}|V_{\bar{1}}) \neq 0$ and $(V_i|V_i) \neq 0$ for at least one $i \in \{\bar{0}, \bar{1}\}$, that is, when the form is neither even nor odd. From a skew-supersymmetric *pre-oscillator form* — an inhomogeneous form whose restrictions to $V_{\bar{0}}$ and $V_{\bar{1}}$ are nondegenerate — they further construct a central extension $V \oplus \mathbb{C}K \oplus \mathbb{C}\kappa$ with an even central element K and an odd central element κ , giving a bracket valued in K when the parities of a generator pair agree and in κ when they differ ([1, Section 4]). Among the relations of (2.1), $[b_u, b_v] = J_{uv}1$ corresponds to the homogeneous Weyl-type commutation relation that also arises from an ordinary (even) bilinear form with no mixed block, while $[b_u, a] = \beta_u \kappa$ is precisely the odd central-valued term added by the mixed block $(b_u|a) \neq 0$; it embodies inhomogeneity in this sense.

Bakalov–Sullivan introduce mixed oscillator brackets and give a trivial abelian extension of $\mathfrak{osp}(1|2)$ in [1, Section 4]. We generalize their deformation, and we prove that the corresponding extension for the family with n oscillator pairs is likewise trivial.

That the extension splits is not unexpected. The simple finite-dimensional Lie superalgebras over $\mathbb{C}$ admitting deformations with odd parameters are classified in [3, Theorem 7.1], the only ones being $\mathfrak{svect}(0|2n+1)$; and an abelian extension of a Lie superalgebra $\mathfrak{s}$ by its adjoint module with parities reversed is classified by the odd part of $H^2(\mathfrak{s}; \mathfrak{s})$. Over $\mathbb{C}$ — the setting Corollary 4.2 specializes to — a splitting therefore exists for $\mathfrak{osp}(1|2n)$ already.

The present theorem is a construction rather than a vanishing statement, and it differs from that in three respects. The vanishing of a class exhibits no primitive: we give f_β explicitly, and with it an even change of generators that is an exact isomorphism. The statement is over the universal ring $R = P \oplus \kappa P$ with the β_u indeterminate. And Γ_β is recovered from a faithful oscillator realization rather than postulated, so the theorem concerns the coefficient the source actually produces — which no statement about cohomology classes addresses.

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Under a general inhomogeneous bilinear form, there is some freedom in how the parameters β_u enter and how the generators are symmetrized. We take the symmetric products $\widehat{L}_{uv}, \widehat{F}_u$ of (2.2) in the Weyl-algebra generators b_u and the odd generator a , fixing how β_u and κ enter so that they satisfy the relations of (2.1). This symmetrization is chosen so that, for $n = 1$, it agrees with the normalization of [1, Section 4] (see Section 5), guaranteeing that the extension to general rank n recovers it as a specialization.

We fix the fermionic square, take even coefficients in the mixed commutators, and use symmetrized oscillator lifts. The odd exterior parameter records the deformation, while the even central generator is specialized to the unit. These choices specify the family under consideration and the coefficient algebra throughout the argument.

The argument is self-contained: we include the ordered basis of the Weyl algebra, the source isomorphism, independence of the lifts, bracket recovery and the coboundary calculation. It is uniform in the number of oscillator pairs.

2 Sources and normalized generators

Fix $n \geq 1$. Put

$$P = \mathbb{Q}[\beta_1, \dots, \beta_{2n}], \quad R = P \oplus \kappa P = P \otimes_{\mathbb{Q}} \Lambda_{\mathbb{Q}}(\kappa), \quad \kappa^2 = 0.$$

All β_u are even and κ is odd. Thus R is a supercommutative algebra. Coefficients are written on the left. For homogeneous elements, writing $p(\cdot) \in \{0, 1\}$ for the parity, we use $[X, Y] = XY - (-1)^{p(X)p(Y)}YX$; braces denote this bracket when both arguments are odd. Let

$$J = \text{diag}(J_2, \dots, J_2), \quad J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Indices run from 1 to $2n$ unless otherwise stated.

Let A_{β} be the unital associative R -superalgebra generated by even b_u and odd a , subject to

$$[b_u, b_v] = J_{uv}1, \quad [b_u, a] = \beta_u \kappa, \quad a^2 = \tfrac{1}{2}1. \quad (2.1)$$

The scalar algebra is supercentral: in particular $a\kappa = -\kappa a$ and $b_u\kappa = \kappa b_u$. Define its symmetrized elements

$$\widehat{L}_{uv} = \tfrac{1}{4}(b_u b_v + b_v b_u) = \widehat{L}_{vu}, \quad \widehat{F}_u = \tfrac{1}{4}(a b_u + b_u a). \quad (2.2)$$

The hats distinguish actual source elements from the coordinate symbols used below.

Remark 2.1 (Even-central specialization). Our constants in (2.1) impose $K = 1$ in an oscillator presentation with even central generator K . This is specialization by the ideal generated by $K - 1$, not omission of scalars. For example, before specialization, the single-pair definitions $H = (b_2 b_1 + b_1 b_2)/4$ and $E^+ = b_2^2/2$ give $[H, E^+] = K E^+$, not E^+ . All normalized tables in this paper are stated after specialization, consistent with the unit-central Fock normalization in [1, Section 4]. The independent odd parameter κ , and terms such as κH , are retained.

Let W_n be the rational Weyl algebra on even B_u , with $[B_u, B_v] = J_{uv}1$. Let

$$C = \mathbb{Q}[a]/(a^2 - \tfrac{1}{2}), \quad A_0 = W_n \otimes_{\mathbb{Q}} C, \quad A_B = R \otimes_{\mathbb{Q}} A_0.$$

Here a is the same odd generator introduced above, and tensor products use graded signs. The algebra C has basis $1, a$, with $a^2 = \tfrac{1}{2}$. Since W_n is purely even, $a B_u = B_u a$ in A_0 and A_B .

Lemma 2.2 (Untwisting the source). *The following are mutually inverse even R -algebra maps, fixing a and R :*

$$\Phi : A_\beta \longrightarrow A_B, \quad b_u \longmapsto B_u + \beta_u \kappa a, \quad \Psi : A_B \longrightarrow A_\beta, \quad B_u \longmapsto b_u - \beta_u \kappa a.$$

Proof. Set $c = \kappa a$. In both presentations, c is even, $c^2 = 0$ and $[c, a] = \kappa$. In A_B , $[B_u, c] = 0$; in A_β , $[b_u, c] = \kappa[b_u, a] = \beta_u \kappa^2 = 0$. The shifted B_u 's therefore retain their Weyl commutators and have mixed commutators $\beta_u \kappa$; the shifted b_u 's retain their Weyl commutators and commute with a . The square and scalar-supercommutation relations are also preserved. Thus both assignments induce homomorphisms. They fix c , so their shifts cancel on every generator in both compositions. $\square$

Write

$$L_{uv}^0 = \frac{1}{4}(B_u B_v + B_v B_u), \quad F_u^0 = \frac{1}{2}a B_u$$

in A_0 . Let $\mathfrak{g}$ be the rational superspace with even basis L_{uv} , $u \leq v$, and odd basis F_u , extending the notation by $L_{vu} = L_{uv}$.

Lemma 2.3 (The undeformed algebra). *The map $\iota_0 : \mathfrak{g} \rightarrow A_0$ defined by $L_{uv} \mapsto L_{uv}^0$, $F_u \mapsto F_u^0$ is injective, with image closed under the supercommutator. The induced bracket is*

$$\{F_u, F_v\}_0 = \frac{1}{2}L_{uv}, \tag{2.3}$$

$$[L_{uv}, F_w]_0 = \frac{1}{2}(J_{vw}F_u + J_{uw}F_v), \tag{2.4}$$

$$[L_{uv}, L_{wz}]_0 = \frac{1}{2}(J_{vw}L_{uz} + J_{uw}L_{vz} + J_{vz}L_{uw} + J_{uz}L_{vw}). \tag{2.5}$$

Proof. For one pair, with $[y, x] = 1$, the monomials $x^i y^j$ span, because any word can be reordered by $yx = xy + 1$; they are independent because $x \mapsto (\text{multiplication by } x)$, $y \mapsto \partial_x$ represents the algebra on $\mathbb{Q}[x]$, and there $x^i y^j$ sends x^j to $j! x^i$ and x^m to 0 for $m < j$, so applying a vanishing combination $\sum c_{ij} x^i y^j$ to $x^0, x^1, \dots$ in turn gives every $c_{ij} = 0$. The n commuting pairs form the tensor product of those one-pair algebras, so their ordered monomials form a basis of W_n . The associated graded algebra of its degree filtration is a commutative polynomial algebra. The leading symbols of the L_{uv}^0 , $u \leq v$, are the distinct quadratic monomials $X_u X_v / 2$; they are independent. The elements $F_u^0 = \frac{1}{2}a B_u$ are independent in the separate a -component of A_0 . This proves injectivity, which also holds after extension to P and R .

Ordinary commutator derivation gives $[L_{uv}^0, B_w] = (J_{vw}B_u + J_{uw}B_v)/2$, proving (2.4). Applying it to each product in L_{wz}^0 gives (2.5). Finally,

$$\{F_u^0, F_v^0\} = \frac{1}{4}a^2(B_u B_v + B_v B_u) = \frac{1}{8}(B_u B_v + B_v B_u) = \frac{1}{2}L_{uv}^0.$$

These identities include repeated indices. This proves closure, and super-Jacobi follows from the associative realization. $\square$

This normalization identifies $\mathfrak{g}$ with the split orthosymplectic algebra. Let $E = \mathbb{Q}^{2n}$ with basis e_u , and let ω be the bilinear form with $\omega(e_u, e_v) = J_{uv}$. Give $M = \mathbb{Q}e_0 \oplus E$ parity $p(e_0) = 0$, $p(E) = 1$, and let $\mathcal{B}$ be the supersymmetric bilinear form with $\mathcal{B}(e_0, e_0) = -1$, $\mathcal{B}|_E = \omega$ and mixed values zero. We then set

$$\text{osp}(M, \mathcal{B}) := \{A \in \text{End}(M) \text{ (homogeneous)} : \mathcal{B}(Ax, y) + (-1)^{p(A)p(x)}\mathcal{B}(x, Ay) = 0 \text{ } (\forall x, y)\},$$

a Lie superalgebra under the supercommutator.

Define its even maps D_{uv} and odd maps q_u by

$$\begin{aligned} D_{uv}(e_0) &= 0, & D_{uv}(z) &= \frac{1}{2}(\omega(e_v, z)e_u + \omega(e_u, z)e_v), \\ q_u(e_0) &= \frac{1}{2}e_u, & q_u(z) &= \frac{1}{2}\omega(e_u, z)e_0 \quad (z \in E). \end{aligned}$$

They lie in $\text{osp}(M, \mathcal{B})$. Every even element of $\text{osp}(M, \mathcal{B})$ vanishes on e_0 and restricts to a symplectic endomorphism of E . The correspondence $A \mapsto ((z, w) \mapsto \omega(Az, w))$ identifies those endomorphisms with symmetric bilinear forms. Nondegeneracy of ω and invertibility of 2 in $\mathbb{Q}$ then show that the D_{uv} , $u \leq v$, are a basis. Every odd element of $\text{osp}(M, \mathcal{B})$ is determined by its value at e_0 and is a unique combination of the q_u . Direct compositions give exactly (2.3)–(2.5) under $L_{uv} \mapsto D_{uv}$, $F_u \mapsto q_u$. For the even-even calculation one may use $[A, D_{wz}] = D_{Ae_w, e_z} + D_{e_w, Ae_z}$, where $D_{x,y}$ for $x, y \in E$ is defined by extending $(e_u, e_v) \mapsto D_{uv}$ bilinearly. This proves a bijective bracket-preserving map from $\mathfrak{g}$ to $\text{osp}(M, \mathcal{B})$, that is, an isomorphism of Lie superalgebras. Multiplying the form by -1 and changing e_{2j-1} to $-e_{2j-1}$ gives, without a field extension, the usual form with even entry 1 and odd block J — exactly the standard form defining $\text{osp}(1|2n)$.

We therefore identify $\mathfrak{g}$, via this isomorphism, with $B(0, n) := \text{osp}(1|2n)$.

Frappat–Sciarrino–Sorba [2] likewise give an oscillator realization of $B(0, n) = \text{osp}(1|2n)$ using only n bosonic oscillator pairs $b_k^\pm$ ($1 \leq k \leq n$) ([2, §2.30, p. 220]). Along the Cartan–Weyl basis for the root system $\{\pm\delta_k \pm \delta_l, \pm 2\delta_k, \pm\delta_k\}$, they build the odd root vectors directly from the same bosonic oscillators as $E_{\pm\delta_k} = b_k^\pm/\sqrt{2}$, a different notation and construction from ours, where we introduce a separate odd generator a ($a^2 = \frac{1}{2}$) and give the odd part as a product with a . We use the auxiliary generator a precisely because it is the technique Bakalov–Sullivan adopt in [1] (an independent odd generator a , with $a \mapsto \xi$ in the Fock representation), rather than the smaller, oscillator-only Fock representation of Frappat et al.

For general n , this correspondence can also be written explicitly, in the undeformed algebra A_0 , where $\mathfrak{g}$ is realized. Split the index set $\{1, \dots, 2n\}$ into n pairs $(2k-1, 2k)$, $k = 1, \dots, n$, and identify each pair with Frappat–Sciarrino–Sorba’s k th bosonic oscillator pair $b_k^- = B_{2k-1}$, $b_k^+ = B_{2k}$. From $[B_{2k-1}, B_{2k}] = 1$ and the definition of L_{uv}^0 we get $B_{2k}B_{2k-1} = 2L_{2k-1, 2k}^0 - \frac{1}{2}$, so their Cartan generators agree with

$$H_k = 2(L_{2k-1, 2k}^0 - L_{2k+1, 2k+2}^0) \quad (1 \leq k \leq n-1), \quad H_n = 2L_{2n-1, 2n}^0. \quad (2.6)$$

That is, (2.6) makes H_k exactly twice the difference of the corresponding pair of L_{uv}^0 ’s here (or, for the last pair, twice $L_{2n-1, 2n}^0$ alone); in H_n the two constants $\frac{1}{2}$ cancel. The factor of 2 reflects a difference in the normalization (the scale of the invariant form) used for the Cartan generators; for $n = 1$, $H_1 = 2L_{12}^0 = 2\iota_0(H)$ with H as in (5.1), which is consistent with the correspondence given there. The same computation confirms that the even root vectors also carry this factor of 2, for instance $E_{2\delta_k} = 2L_{2k, 2k}^0$, $E_{-2\delta_k} = 2L_{2k-1, 2k-1}^0$, and $E_{\delta_k - \delta_{k+1}} = 2L_{2k, 2k+1}^0$. The odd root vectors $E_{\delta_k} = B_{2k}/\sqrt{2}$ and $E_{-\delta_k} = B_{2k-1}/\sqrt{2}$, on the other hand, lie outside $\iota_0(\mathfrak{g})$: the odd part of the realization is spanned by the $F_u^0 = \frac{1}{2}aB_u$, and $B_u = 4aF_u^0$, so passing between the two descriptions uses the auxiliary generator a . This is exactly the construction difference noted above: an oscillator-only (smaller) Fock representation versus the auxiliary generator a that follows Bakalov–Sullivan. Since the even Cartan–Weyl structure matches exactly up to this normalization, and the root system $\{\pm\delta_k \pm \delta_l, \pm 2\delta_k, \pm\delta_k\}$ together with the superrank $(n(2n+1)|2n)$ (Proposition 3.1) matches the structure of $\text{osp}(1|2n)$ over $\text{sp}(2n)$, this general- n correspondence corroborates, in terms of Frappat et al.’s standard definition, the abstract isomorphism $\mathfrak{g} \cong \text{osp}(M, \mathcal{B})$ (via D_{uv} , q_u) proved above. In any case, the proof that $\mathfrak{g}$ is isomorphic to $\text{osp}(1|2n)$ relies solely on that construction via D_{uv} , q_u ; the correspondence with Frappat et al.’s notation is a supplementary, corroborating remark.

3 Recovery of the deformed bracket

The term *recovery* is used below in the following sense. Starting from the (anti)commutators that actually hold in the concrete realization A_β (relations (2.1)), we induce a new bracket $[\cdot, \cdot]_\beta$ on the coordinate module $\mathfrak{g}_P$ (and its extension $\mathfrak{g}_R = \mathfrak{g}_P \oplus \kappa \mathfrak{g}_P$ by the odd parameter κ) by requiring the source map ι_β to preserve the bracket, and we read off, from the side of the realization, what explicit closed form this bracket takes. The direction runs one way, from the realization (the generators and relations) to the coordinates (the structure constants). The content of Proposition 3.1 is that, in this sense, the bracket induced by ι_β is well defined on the extension $\mathfrak{g}_R$ (consistently, satisfying super-skew symmetry and the super-Jacobi identity) and is given by the explicit closed form (3.2).

Set $\mathfrak{g}_P = P \otimes_{\mathbb{Q}} \mathfrak{g}$ and $\mathfrak{g}_R = R \otimes_{\mathbb{Q}} \mathfrak{g}$, and extend the base bracket, for homogeneous elements r, s of P or R and homogeneous basis elements X, Y of $\mathfrak{g}$, by

$$[rX, sY]_0 = (-1)^{p(X)p(s)} rs[X, Y]_0. \quad (3.1)$$

The source map $\iota_\beta : \mathfrak{g}_R \rightarrow A_\beta$ sends coordinate basis elements to (2.2) and is even R -linear. Its injectivity and closed image follow from the next proposition.

Proposition 3.1 (Faithful recovery). *The map ι_β is injective with closed image. The recovered bracket on coordinate elements $X, Y \in \mathfrak{g}_P$ is*

$$[X, Y]_\beta = [X, Y]_0 + \kappa \Gamma_\beta(X, Y), \quad (3.2)$$

where the odd P -bilinear coefficient is determined by

$$\Gamma_\beta(L_{uv}, L_{wz}) = 0, \quad (3.3)$$

$$\Gamma_\beta(L_{uv}, F_w) = \frac{1}{2}(\beta_u L_{vw} + \beta_v L_{uw}), \quad (3.4)$$

$$\Gamma_\beta(F_u, F_v) = -\frac{1}{2}(\beta_u F_v + \beta_v F_u). \quad (3.5)$$

The bracket on $\mathfrak{g}_R$ uses the even scalar rule (3.1). Its free R -superrank is $(n(2n+1)|2n)$.

Proof. Lemma 2.2 gives

$$\Phi(\widehat{F}_u) = F_u^0, \quad \Phi(\widehat{L}_{uv}) = L_{uv}^0 + \kappa(\beta_u F_v^0 + \beta_v F_u^0). \quad (3.6)$$

For the first equality the two correction terms cancel because $a\kappa a = -\kappa/2$ and $\kappa a^2 = \kappa/2$. The second follows by expanding the two products and using $(\kappa a)^2 = 0$.

Put $h(L_{uv}) = \beta_u F_v + \beta_v F_u$, $h(F_u) = 0$, as an odd P -linear map. Define the even R -linear map $U : \mathfrak{g}_R \rightarrow \mathfrak{g}_R$ on the basis by

$$U(L_{uv}) = L_{uv} + \kappa h(L_{uv}), \quad U(F_u) = F_u.$$

It is invertible, with the same basis formula and a minus sign in the correction. Equation (3.6) is $\Phi \iota_\beta = \iota_{0,R} U$, where $\iota_{0,R} : \mathfrak{g}_R \rightarrow A_B$ is the even R -linear extension of ι_0 . Hence ι_β is injective and its image is carried onto the entire closed image of $\iota_{0,R}$. The recovered bracket is therefore, before introducing a coboundary,

$$[X, Y]_\beta = U^{-1}[UX, UY]_0. \quad (3.7)$$

For the even-even sector the required cancellation is

$$h([L, L']_0) = [h(L), L']_0 + [L, h(L')]_0 \quad (3.8)$$

for even L, L' . Indeed, set

$$v = \sum_{j=1}^n (\beta_{2j} e_{2j-1} - \beta_{2j-1} e_{2j}), \quad (3.9)$$

so that $\sum_t v_t J_{tu} = \beta_u$, and write $F(v) = \sum_t v_t F_t$. Equation (2.4) gives $[2F(v), L_{uv}]_0 = h(L_{uv})$. Super-Jacobi on two even inputs proves (3.8). Expansion of (3.7) now proves (3.3). For an even-odd pair, its correction is $[h(L_{uv}), F_w]_0 = (\beta_u L_{vw} + \beta_v L_{uw})/2$. For an odd-odd pair, U fixes both inputs, so the correction is $-h(L_{uv})/2$. This proves the remaining formulas, also when indices repeat.

Finally, $\mathfrak{g}_R = \mathfrak{g}_P \oplus \kappa \mathfrak{g}_P$, so the coefficient in (3.2) is unique as an element of $\mathfrak{g}_P$. Counting the independent lifts gives the superrank. $\square$

4 An explicit odd coboundary

For an odd P -linear cochain $f : \mathfrak{g}_P \rightarrow \mathfrak{g}_P$, our coboundary convention is

$$(\delta f)(X, Y) = [f(X), Y]_0 + (-1)^{p(X)} [X, f(Y)]_0 - f([X, Y]_0). \quad (4.1)$$

By super-skew symmetry, the first term equals $-(-1)^{(p(X)+1)p(Y)} [Y, f(X)]_0$. Thus (4.1) agrees with [1, Equation (4.5)]. All cochains here are P -linear and do not differentiate parameter polynomials. We call the family trivial when its recovered coefficient is such a coboundary; the theorem also gives the corresponding exact even isomorphism.

Theorem 4.1 (Family triviality for every rank). *For every $n \geq 1$, the coefficient in Proposition 3.1 satisfies $\Gamma_\beta = \delta f_\beta$, where*

$$f_\beta(L_{uv}) = 0, \quad f_\beta(F_u) = \sum_{j=1}^n (\beta_{2j-1} L_{2j,u} - \beta_{2j} L_{2j-1,u}). \quad (4.2)$$

Extend f_β as an odd map by $(f_\beta)_R(rX) = (-1)^{p(r)} r f_\beta(X)$. Then

$$T_\beta = \text{id} + \kappa(f_\beta)_R, \quad T_\beta^{-1} = \text{id} - \kappa(f_\beta)_R \quad (4.3)$$

are even R -linear inverse maps, and

$$[T_\beta X, T_\beta Y]_0 = T_\beta([X, Y]_\beta). \quad (4.4)$$

Proof. With v from (3.9), write $f_\beta(F_u) = -\sum_t v_t L_{tu}$. The even-even coboundary is zero because f_β vanishes on the even part. For the even-odd sector, (2.5) and $\sum_t v_t J_{ut} = -\beta_u$ give

$$\begin{aligned} [L_{uv}, f_\beta(F_w)]_0 &= \frac{1}{2}(\beta_v L_{uw} + \beta_u L_{vw}) \\ &\quad + \frac{1}{2}(J_{vw} f_\beta(F_u) + J_{uw} f_\beta(F_v)). \end{aligned}$$

The second line is precisely $f_\beta([L_{uv}, F_w]_0)$; subtracting it proves (3.4). For two odd inputs,

$$[f_\beta(F_u), F_v]_0 = -\frac{1}{2}(J_{uv} F(v) + \beta_v F_u).$$

Interchanging u, v and adding cancels the J -terms. The image $f_\beta(\{F_u, F_v\}_0)$ is zero, yielding (3.5). This proves $\delta f_\beta = \Gamma_\beta$ in all parity sectors, including the diagonal odd rows.

Set $D = \kappa(f_\beta)_R$. It is even and R -linear, and $D^2 = 0$ by the odd scalar-extension rule and $\kappa^2 = 0$. For homogeneous $X, Y \in \mathfrak{g}_P$, expansion gives

$$\begin{aligned} [T_\beta X, T_\beta Y]_0 &= [X, Y]_0 + \kappa([f_\beta(X), Y]_0 + (-1)^{p(X)}[X, f_\beta(Y)]_0), \\ T_\beta([X, Y]_\beta) &= [X, Y]_0 + \kappa(f_\beta([X, Y]_0) + \Gamma_\beta(X, Y)). \end{aligned}$$

They agree by (4.1); even R -bilinearity extends the identity to all inputs. The inverse in (4.3) follows from $D^2 = 0$. $\square$

The two maps U and T_β need not coincide. Indeed, on $\mathfrak{g}_P$ one has $f_\beta = h - \text{ad}(2F(v))$: on the even part this follows from (3.8)'s proof, and on F_u it follows from $[2F(v), F_u]_0 = \sum_t v_t L_{tu}$. This only uses the elementary fact that inner maps are derivations, not a classification of all homogeneous cocycles.

Corollary 4.2 (Complex parameters). *For every tuple $\lambda \in \mathbb{C}^{2n}$, specialize $\beta_u \mapsto \lambda_u$ and retain the odd exterior parameter κ . The specialized lifts are faithful, and (4.2)–(4.4) define a trivialization over $\Lambda_{\mathbb{C}}(\kappa)$.*

Proof. The polynomial identities and the explicit inverse maps Φ, Ψ specialize, because specialization is a ring map. Since $P \rightarrow \mathbb{C}$ is a quotient, faithfulness needs its own argument: the complex Weyl algebra has the same ordered basis, and the quadratic leading symbols and the odd linear monomials in Lemma 2.3 remain independent. Thus the proof of Proposition 3.1 applies after specialization. $\square$

5 The reference example and remarks

For $n = 1$, identify

$$H = L_{12}, \quad E^+ = L_{22}, \quad E^- = -L_{11}, \quad F^+ = F_2, \quad F^- = F_1. \quad (5.1)$$

The source elements are the symmetrized expressions of [1, Equation (4.3)], with the specialization stated in Remark 2.1. At $\beta_1 = 1, \beta_2 = 0$, the source relations give $[b_1, a] = \kappa$, $[b_2, a] = 0$, and our recovered table becomes

$$\begin{aligned} [H, E^\pm]_\beta &= \pm E^\pm, & [E^+, E^-]_\beta &= 2H, \\ [H, F^+]_\beta &= \frac{1}{2}F^+ + \frac{\kappa}{2}E^+, & [H, F^-]_\beta &= -\frac{1}{2}F^- + \frac{\kappa}{2}H, \\ [E^+, F^-]_\beta &= -F^+, & [E^-, F^+]_\beta &= -F^- - \kappa H, \\ [E^-, F^-]_\beta &= \kappa E^-, & \{F^+, F^-\}_\beta &= \frac{1}{2}H - \frac{\kappa}{2}F^+, \\ \{F^+, F^+\}_\beta &= \frac{1}{2}E^+, & \{F^-, F^-\}_\beta &= -\frac{1}{2}E^- - \kappa F^-. \end{aligned}$$

This agrees with [1, Equation (4.4)]; the other raising odd bracket is $[E^+, F^+]_\beta = 0$. The primitive specializes to

$$f_\beta(H) = f_\beta(E^\pm) = 0, \quad f_\beta(F^+) = E^+, \quad f_\beta(F^-) = H,$$

recovering the solution stated after [1, Equation (4.5)]. Theorem 4.1 includes this example and applies to every $n \geq 1$.

By (3.5), $\Gamma_\beta(F_u, F_u) = -\beta_u F_u$. Over the universal ring, and after any fixed rational or complex specialization of the parameters, independence of the F_u therefore makes Γ_β vanish only when every β_u does, while its class vanishes for every choice of the β_u .

Fock module. Let $\mathcal{F} = R[x_1, \dots, x_n] \otimes_{\mathbb{Q}} C$, free of rank two over $R[x_1, \dots, x_n]$ on the basis $1, a$ of C , with the x_k even. Sending

$$B_{2k-1} \mapsto \partial_{x_k}, \quad B_{2k} \mapsto x_k$$

and letting a act by left multiplication represents A_B on $\mathcal{F}$; composing with Φ of Lemma 2.2 represents A_β by

$$b_u \mapsto B_u + \beta_u \kappa a, \quad a \mapsto a.$$

The relations (2.1) hold because κa is even and commutes with the B_u , $(\kappa a)^2 = -\kappa^2 a^2 = 0$, and $[\kappa a, a] = \kappa$. Both representations are faithful: W_n acts faithfully on $R[x_1, \dots, x_n]$ in characteristic zero and C acts faithfully on itself, and Φ is an isomorphism. This gives a second route to the source faithfulness used in Corollary 4.2. The element annihilated by every b_{2k-1} is $1 - \sum_k \beta_{2k-1} x_k \kappa a$. For $n = 1$ with $\beta_1 = 1$, $\beta_2 = 0$, and after the specialization of Corollary 4.2 has made the base the field $\mathbb{C}$, $\mathcal{F}$ is the Fock space of the last paragraph of [1, Section 4], up to the choice of operator realizing the mixed term: there it is realized by an odd derivation, and the annihilated element is 1.

The family treated here is the $m = 0$ member of the orthosymplectic series: $B(m, n)$ adds $2m$ fermionic oscillators to the same $2n$ bosonic ones. In Frappat–Sciarrino–Sorba’s realization of $B(m, n)$ the odd root vectors along $\pm \delta_k$ are products $b_k^\pm e$ with a supplementary odd generator e , $e^2 = 1$ — the role played here by a , normalized to $a^2 = \frac{1}{2}$ after [1] rather than $e^2 = 1$ — whereas their $B(0, n)$, treated separately, dispenses with e and realizes those vectors by the bosonic oscillators alone ([2, §2.30, pp. 219–220]), as recalled in Section 2. We keep the generator a , following [1], because the deformation acts on it: $[b_u, a] = \beta_u \kappa$ in (2.1). The central extension of [1, Section 4] is constructed from an arbitrary skew-supersymmetric pre-oscillator form; [1] classifies the irreducible such forms up to dimension seven. The form used in their Section 4 example is (2.1) at $n = 1$; what is treated here is its extension to every rank.

A Machine-checked formalization

The results stated above have been formalized and machine-checked in the Lean 4 proof assistant, with the exceptions recorded in Section A.4. This appendix states exactly which assertions are checked, names the declaration that carries each one, and is explicit about what the formalization does not establish. Nothing in the body of the paper depends on this appendix.

At the time of writing, Mathlib has no library for Lie superalgebras. The formalization therefore builds the necessary structure from scratch, including the graded bracket, parity, and the scalar rule. That structure is stated for a free module on an indexed basis and is uniform in the rank, so it is tied to the presentation rather than to this family: it carries any Lie superalgebra given by structure constants on such a basis.

A.1 The formalized objects

The formalization works with the coordinate presentation directly. For each $n \geq 1$ it fixes the free module over $P = \mathbb{Q}[\beta_1, \dots, \beta_{2n}]$ on the indexed basis $\{L_{uv}\} \cup \{F_u\}$, with the parity of Section 2, and defines the undeformed bracket by the structure constants of Lemma 2.3. This is the object called $\mathfrak{g}_P$ above. The matrix J is defined by the same parity-and-successor rule, not tabulated, so every statement below is uniform in n ; no assertion is proved by enumeration at a fixed rank.

The coefficient Γ_β is defined by (3.3)–(3.5) and the primitive f_β by (4.2), both transcribed from their statements above rather than derived. Below, (L, L) , (L, F) , (F, F) and (F, L) name the four

sectors of Γ_β by the kinds of basis element substituted into its two arguments, in that order; so (F, L) is $\Gamma_\beta(F_u, L_{vw})$. That the (F, L) value carried by the definition is the only one the source admits is item 5 below. The module $\mathfrak{g}_R = \mathfrak{g}_P \oplus \kappa \mathfrak{g}_P$ is presented as the free P -module on two copies of that basis, the second carrying the parity shift of the odd parameter κ , with the bracket extended by the scalar rule (3.1). Section A.4 records what that presentation does and does not assume.

A.2 What is proved

For every $n \geq 1$, with no restriction on the rank and no hypothesis beyond those displayed:

1. **The coordinate presentation is a Lie superalgebra of the stated shape.** The bracket respects the $\mathbb{Z}/2$ -grading, is super-skew symmetric, and satisfies the super-Jacobi identity, at the level of basis elements and for arbitrary homogeneous module elements.
2. **The deformation coefficient is a coboundary:** $\Gamma_\beta = \delta f_\beta$ on $\mathfrak{g}_P$, with δ as in (4.1). This is Theorem 4.1's first assertion for the Γ_β of (3.3)–(3.5).
3. **The change of generators is an exact isomorphism.** T_β and T_β^{-1} of (4.3) are even, R -linear, and mutually inverse exactly, and they satisfy the intertwining (4.4). The intertwining is proved for arbitrary elements of $\mathfrak{g}_R$, with no homogeneity hypothesis.
4. **The source recovery holds.** Proposition 3.1 is proved for the deformed source as realized below: ι_β is injective, the bracket it recovers on coordinate elements is $[X, Y]_0 + \kappa \Gamma_\beta(X, Y)$ with Γ_β the coefficient of (3.3)–(3.5), that coefficient is unique as an element of $\mathfrak{g}_P$, and the image of ι_β is closed under the bracket. The recovered bracket is obtained as (3.7) states it, by conjugating with U .
5. **The fourth sector is forced.** The concrete algebra admits exactly one coefficient on the (F, L) sector, namely $-\frac{1}{2}(\beta_u L_{vw} + \beta_v L_{uw})$, which is the value (3.4) and super-skew symmetry give. Section A.4 records what this does and does not settle.

Items 2 and 3 together are the content of Theorem 4.1. By item 4 the coefficient they concern is the one recovered from the oscillator source, so no identification is left to the reader: Theorem 4.1 is checked as stated in Section 4, not only for the coefficient written down in (3.3)–(3.5).

At $n = 1$, and for one explicitly given datum only, the formalization also checks that an external description of the bracket table decodes to exactly the bracket defined internally. This is a consistency check on a single object, not a statement about a format or a program.

A.3 Declarations

Each assertion above is carried by the declarations named here. The names are those of the accompanying development, and each is a link to the module holding it in the repository [4] at the tag `v2-lean-formalization`; the identities pinning the exact version are in Section A.5.

Item 1 (the coordinate presentation). Grading is `bracketN_isHomogN` and super-skew symmetry is `bracketN_super_skew_homog`, both in `IndexedLaws`; the super-Jacobi identity is `jacobiN_homog`, in `IndexedJacobi`.

Item 2 (the coboundary identity). The identity as stated is `GammaBetaN_eq_deltaFN`, and `G4` is the same identity at the level of basis elements; item 5 records what separates the two. Both in `IndexedCoboundary`.

Item 3 (the change of generators). `TBetaInv_TBeta` and `TBeta_TBetaInv` are the two inverse identities, `TBeta_isHomogR` is evenness, and `intertwining` is (4.4). These are in `IndexedKappa`, as are the four checks to which item 4 holds the presentation of $\mathfrak{g}_R$ accountable: that the bracket restricts to the undeformed one on $\mathfrak{g}_P$ (`bracketR_iotaR_iotaR`), that it is super-skew (`bracketR_super_skew_homog`), that $\kappa^2 = 0$ (`kappaMulR_sq`), and that multiplication by κ is not injective (`kappaMulR_not_injective`). The corresponding restriction for the deformed bracket is `bracketRBeta_iotaR_iotaR`.

Item 4 (source recovery). `iotaBetaR_injective` is injectivity of ι_β , `prop_recovery` is the recovered bracket (3.2), and `coefficient_unique` is uniqueness of the coefficient as an element of $\mathfrak{g}_P$; these are in `SourceRecoveryClosure`. Closure of the image is `iotaBetaR_bracket_closed`, in `SourceRecoveryClosedImage`.

Item 5 (the (F, L) sector is forced). `GammaBetaBasis_Fof_Lof_forced_FL_free`, in `SourceRecoveryFLConvention`.

The $n = 1$ consistency check of Section A.2's closing paragraph is `decodedBracket_eq_bracket`, in `Bridge`.

A.4 What is not formalized

The following are stated so that no reader infers more from this appendix than it supports. They are not conjectures or work in progress; they are the boundary of what has been checked.

1. **The source is realized, not presented abstractly.** Item 4 proves Proposition 3.1 for a concrete realization of A_β : the development exhibits generators inside an explicit algebra and proves that they satisfy (2.1), rather than constructing the algebra presented by those relations and identifying it. Lemma 2.2 as an abstract statement is therefore not formalized; nor are Corollary 4.2 and the identity $f_\beta = h - \text{ad}(2F(v))$ of Section 4.
2. **The development does not follow the proof given here line by line.** Section 4 derives Γ_β from the source and then exhibits the primitive; the formalization posits the coordinate formulas, proves the coboundary identity directly, and recovers the same coefficient from the source separately (item 4). In particular, it does not construct the isomorphism Φ of Lemma 2.2: both algebras are realized in one ambient algebra, so the change of lifts (3.6) is a computation there. The conclusions agree; the route does not, and a reader should not take the formalization as a transcription of this paper's argument.
3. **The (F, L) value is carried as a definition.** Equations (3.3)–(3.5) specify Γ_β on (L, L) , (L, F) and (F, F) ; the development fixes the (F, L) value by super-skew symmetry. Item 5 shows the source admits no other value there, so that definition is not an independent input. It remains a definition: the declaration equating the (F, L) row with that value proves it by unfolding the definition, so it records the choice rather than deriving it.
4. **$\mathfrak{g}_R$ is presented, not constructed.** The development does not build R as a supercommutative algebra object and $\mathfrak{g}_R$ as $R \otimes_{\mathbb{Q}} \mathfrak{g}$; it presents $\mathfrak{g}_P \oplus \kappa \mathfrak{g}_P$ as a free P -module on a doubled basis and takes (3.1) as the definition of the bracket. The presentation is held accountable by the four checks listed for it in Section A.3: it restricts to the undeformed bracket on $\mathfrak{g}_P$, it is super-skew, $\kappa^2 = 0$, and multiplication by κ is *not* injective.
5. **δf is the bilinear extension of its values on basis elements.** The sign $(-1)^{p(X)}$ in (4.1) is defined only on homogeneous arguments, so item 2 is the statement that two bilinear extensions agree.

A.5 Reproducing the check

The development builds against Lean 4 toolchain `leanprover/lean4:v4.29.1` and `mathlib` at commit `5e932f97dd25535344f80f9dd8da3aab83df0fe6`, both pinned in the accompanying sources. Building the project prints the axiom dependencies of every declaration listed in Section A.3 together with the others it audits: 649 declarations, of which 634 depend on axioms and 15 on none. Every one depends on at most `propext`, `Classical.choice` and `Quot.sound`, the standard axioms of Lean’s logic. **No declaration depends on `sorryAx`**, so no statement is assumed rather than proved, and no additional axiom is introduced. The development uses no `native_decide` and no compiler-trusted evaluation.

The sources accompanying this paper are the repository [4]. It contains the modules named above, the $n = 1$ data instance whose decoding is the last entry of Section A.3’s table, and the pinned toolchain and dependency revisions, so a reader can rebuild and compare rather than rely on this summary. Building it prints the axiom output quoted here.

Use of generative AI

Generative-AI tools assisted with mathematical discussion, checking and drafting. The argument and its classical theorem input are stated explicitly above. Appendix A records a machine-checked formalization of the statements listed there, in the Lean 4 proof assistant, with the exclusions stated in that appendix. No computer-algebra computation is claimed. The author reviewed and edited all of it, and takes full responsibility for the contents of this paper, irrespective of how they were generated.

References

- [1] Bojko Bakalov and McKay Sullivan, *Inhomogeneous supersymmetric bilinear forms*, Contemp. Math. **713**, Amer. Math. Soc., Providence, RI, 2018, pp. 35–45. <https://arxiv.org/abs/1612.09400v2>. Section and equation numbers cited here are those of the arXiv version v2.
- [2] Luc Frappat, Antonino Sciarrino, and Paul Sorba, *Dictionary on Lie Algebras and Superalgebras*, Academic Press, San Diego, CA, 2000.
- [3] Dmitry Leites, *On odd parameters in geometry*, J. Lie Theory **33** (2023), no. 4, 965–1004. <https://arxiv.org/abs/2210.17096>.
- [4] Hisashi Aoi, *osp-triviality: a Lean 4 formalization of the triviality of inhomogeneous deformations of $\mathfrak{osp}(1|2n)$* , GitHub repository, 2026. <https://github.com/ritsumei-aoi/osp-triviality>. The state accompanying this version is the tag `v2-lean-formalization`; the superseded computational state remains at `v1-python-arXiv-2604.05252`.